Advanced RAG & Evaluation

Student Research Topic Guide

Prepared for: ITE / AX Specialize Program — Mastering Local LLM

KSGA — Korea Software HRD Center (KSHRD)

Overview

This guide breaks down two research modules — Advanced RAG & Evaluation — into student-assignable subtopics. Each subtopic includes an estimated research time (reading plus light hands-on practice, not a full build). Estimates assume a student with basic Python background researching independently.

Module 2: Advanced RAG Architecture and Evaluation

1. Hybrid Retrieval: Keyword + Vector Search *— Total: ~2.5 hrs*

- Limitations of pure dense (vector) search: misses exact terms, IDs, acronyms *(20 min)*
- Limitations of pure keyword search (BM25): misses semantic meaning *(20 min)*
- BM25 fundamentals (term frequency, inverse document frequency) *(30 min)x*
- Combining scores: Reciprocal Rank Fusion (RRF) *(30 min)*
- Weighted score fusion vs rank fusion *(20 min)*
- Tools/libraries: Elasticsearch, OpenSearch, Qdrant/Weaviate hybrid modes *(30 min)*

2. Graph RAG *— Total: ~3 hrs*

- Limitations of chunk-based retrieval for multi-hop or relational questions *(20 min)*
- Building a knowledge graph from documents (entities + relationships) *(45 min)*
- Graph construction: entity extraction, relationship extraction *(45 min)*
- Community detection and summarization (e.g., Microsoft GraphRAG approach) *(45 min)*
- Querying: local search (entity-focused) vs global search (theme-focused) *(30 min)*
- Graph databases: Neo4j basics *(40 min)*

3. Agentic RAG *— Total: ~3 hrs*

- What makes RAG "agentic" — reasoning and tool use vs single-pass retrieval *(20 min)*
- Query planning and decomposition (breaking complex questions into sub-queries) *(30 min)*
- Iterative/multi-step retrieval (retrieve → evaluate → retrieve again) *(30 min)*

- Self-reflective RAG (the model critiques its own retrieved context) *(30 min)*
- Tool-calling agents that choose between retrieval, search, or calculation *(30 min)*
- Frameworks: LangGraph, LlamaIndex agents (conceptual overview) *(40 min)*

4. Reranking with Cross-Encoders *— Total: ~2 hrs*

- Bi-encoders (embedding models) vs cross-encoders — architectural difference *(30 min)*
- Why rerankers improve precision after initial retrieval *(20 min)*
- Two-stage retrieval pipeline: retrieve broad (top-50) → rerank to top-5 *(30 min)*
- Popular rerankers: Cohere Rerank, BGE-reranker, cross-encoder/ms-marco models *(25 min)*
- Latency/cost trade-offs of reranking *(15 min)*

5. Context Management *— Total: ~2 hrs*

- Context window budgeting across multiple retrieved chunks *(20 min)*
- Ordering/placement effects ("lost in the middle" phenomenon) *(30 min)*
- Deduplication of overlapping or redundant chunks *(20 min)*
- Summarization of retrieved content before generation *(25 min)*
- Multi-document and long-context strategies *(25 min)*

6. Prompt Compression *— Total: ~1.5 hrs*

- Why compress prompts (cost, latency, context limits) *(15 min)*
- Extractive compression (keeping only key sentences) *(25 min)*
- Abstractive compression (LLM-generated summaries of context) *(25 min)*
- Tools: LLMLingua and similar approaches *(25 min)*
- Trade-offs: compression ratio vs information loss *(10 min)*

7. Debugging Common RAG Failure Modes *— Total: ~2.5 hrs*

- Hallucination despite retrieval (model ignores or contradicts context) *(25 min)*
- Missing context (relevant chunk not retrieved — recall failure) *(25 min)*
- Retrieval mismatch (wrong chunk retrieved due to embedding/query mismatch) *(25 min)*
- Chunking failures (context split across chunk boundaries) *(20 min)*
- Query ambiguity and query rewriting/expansion as a fix *(30 min)*
- Building a debugging checklist / root-cause framework *(25 min)*

8. Guardrails for RAG Systems *— Total: ~2 hrs*

- Input guardrails: query filtering, prompt injection detection *(30 min)*
- Output guardrails: fact-checking generated answers against retrieved context *(25 min)*
- Citation/attribution enforcement (forcing answers to cite sources) *(25 min)*

- Refusal handling when no relevant context is found *(20 min)*
- Tools: Guardrails AI, NeMo Guardrails (conceptual overview) *(20 min)*

9. RAG Evaluation *— Total: ~3 hrs*

- Retrieval metrics: precision@k, recall@k, MRR (Mean Reciprocal Rank) *(30 min)*
- Generation metrics: faithfulness/groundedness, answer relevance *(30 min)*
- Frameworks: RAGAS, TruLens, DeepEval *(40 min)*
- Building a golden dataset (Q&A pairs with known correct answers/sources) *(30 min)*
- Human evaluation vs LLM-as-judge evaluation *(25 min)*
- End-to-end pipeline evaluation vs component-level evaluation *(25 min)*

Module total: ~22 hours